

Tutorial 13

Inner product spaces

1. Let V be an inner product space and let $\mathbf{u}, \mathbf{v} \in V$. For each of the following statements, say whether it is true or false and give a reason for your answer.
 - (a) If $\mathbf{u}$ and $\mathbf{v}$ are linearly independent, they are orthogonal.
 - (b) If $\mathbf{u}$ and $\mathbf{v}$ are orthogonal, they are linearly independent.
 - (c) If $\mathbf{w}$ is orthogonal to both $\mathbf{u}$ and $\mathbf{v}$, it is orthogonal to every vector in $\text{span}(\{\mathbf{u}, \mathbf{v}\})$.

2. For any $p(x), q(x) \in P_n$, put

$$\langle p(x), q(x) \rangle = \int_{-1}^1 p(x)q(x) dx.$$

- (a) Show that this defines an inner product on P_n .
Do the check that $\langle p(x), p(x) \rangle = 0 \Rightarrow p(x) = \mathbf{0}$ for the case $n = 1$; for larger n it gets a bit messy, but the idea is the same.
- (b) Let $n = 3$ and $p(x) = x^3 - 1$. Find $\|p(x)\|$.
- (c) Let $p(x) = x^3 - 1$ and $q(x) = x^3 + 1$. Determine $\|q(x)\|$, $\langle p(x), q(x) \rangle$, and the angle between $p(x)$ and $q(x)$. The *angle* between $p(x)$ and $q(x)$ is defined to be θ , where

$$\cos \theta = \frac{\langle p(x), q(x) \rangle}{\|p(x)\| \|q(x)\|}.$$

3. Let V be an inner product space and suppose that $\mathbf{u}, \mathbf{v}$ and $\mathbf{w}$ are vectors such that

$$\langle \mathbf{u}, \mathbf{v} \rangle = 2, \quad \langle \mathbf{v}, \mathbf{w} \rangle = -3, \quad \langle \mathbf{u}, \mathbf{w} \rangle = 5, \quad \|\mathbf{u}\| = 1, \quad \|\mathbf{v}\| = 2, \quad \|\mathbf{w}\| = 7$$

Evaluate each of the following using this information:

- (a) $\langle \mathbf{u} + \mathbf{v}, \mathbf{v} + \mathbf{w} \rangle$
- (b) $\langle \mathbf{u} - \mathbf{v} - 2\mathbf{w}, 4\mathbf{u} + \mathbf{v} \rangle$
- (c) $\|2\mathbf{w} - \mathbf{v}\|$.

4. Let

$$A = \left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} \right\} \text{ and } C = A \cup \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \end{pmatrix} \right\}.$$

- (a) Show that A is an orthogonal set in $\mathbb{R}^4$, with its usual inner product. Is it an orthonormal set?
- (b) Use the Gram-Schmidt procedure to find an orthonormal basis B for $\text{span}(C)$.

5. Let A be a non-empty subset of an inner product space V . Define

$$A^\perp = \{ \mathbf{v} \in V : \langle \mathbf{v}, \mathbf{u} \rangle = 0 \text{ for all } \mathbf{u} \in A \}.$$

- (a) We start with an example. Let $V = \mathbb{R}^3$, with its usual inner product. Find $A^\perp$ if
 - (i) $A = \{(0, 0, 1)\}$,
 - (ii) $A = \{(1, 0, 0), (0, 0, 1)\}$.
- (b) In general (i.e. not only in the cases looked at in (a)), is $A^\perp$ always a subspace of V ? Justify your answer.
- (c) Is it possible to have $A^\perp = V$? If so, when will this happen?
- (d) If U is a **subspace** of V , find $U \cap U^\perp$.

6. Let $C = \begin{pmatrix} 4 & -1 \\ -1 & 4 \end{pmatrix}$ and for $\mathbf{u} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$, $\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} \in \mathbb{R}^2$ define $\langle \mathbf{u}, \mathbf{v} \rangle$ by the matrix product

$$\langle \mathbf{u}, \mathbf{v} \rangle = (u_1 \ u_2) C \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}.$$

- (a) Show that this defines an inner product on $\mathbb{R}^2$. [Hint: Do the checks of conditions 2 and 3 in the definition (those involving addition and scalar multiplication) *without* multiplying out the righthand side; this goes more quickly.]
Use this inner product (not the usual one on $\mathbb{R}^2$) to answer the next two questions

(b) Find $\left\| \begin{pmatrix} 4 \\ 1 \end{pmatrix} \right\|$.

- (c) Find all vectors in $\mathbb{R}^2$ that are orthogonal (perpendicular) to $\begin{pmatrix} 4 \\ 1 \end{pmatrix}$ (i.e. those whose inner product with this vector gives 0).
- (d) (A bit of a challenge) Can we replace the matrix C by any non-zero 2×2 matrix and still get an inner product? Give a reason for your answer.